June 2, 1958

CHECK LIST FOR SIMPLIFYING SIGMA

1. Eliminate the Hamming checker-generator from the computer.

   Possible saving of transistors (2400).
   Possibility of improving Sigma performance by reducing the cycle time of the I box.

2. * Eliminate decimal multiply, divide, and cumulative multiply.

   Possible saving of transistors 2500.
   "ADD" — 550

3. Eliminate the ability to position full word instructions across word boundaries in program executions.

   Possible saving of transistors 200.

4. Eliminate the ability to freely attach 0.5 us memories to the Sigma computer system.

   Possible saving of transistors _____.

5. Eliminate the use of algebraic signs in Index Arithmetic.

   Possible saving of transistors 400.
   Possibility of contributing toward improving Sigma performance by reducing the cycle time of the I box.

6. Eliminate Progressive Indexing.

   Possible saving of transistors 300.

7. * Eliminate the Left Zero Counter and All Ones Counter.

   Possible savings of transistors 2500.

8. * Eliminate the Address Comparison feature for multi-program zone protection.

   Possible saving of transistors 2700.

9. * Eliminate the Elapsed Time Clock

   Possible saving of transistors 200.

CHECK LIST FOR SIMPLIFYING SIGMA (cont.)

10. Restrict VFL operations to bytes of 4 or 8 only, and further specify that all boundary positions must be modulo 4.

Possible saving of transistors _____.

11. Reduce the level of completeness of checking that is now provided.

Possible saving of transistors 0 - 7000.

12. Eliminate the use of an adder to generate an interrupt address, specifying using the prefix technique.

Possible saving of transistors 0.

13a. Simplify instruction definitions and reduce the number of different useable formats.

13b. Eliminate immediate Transmits.

13c. Simplify Branch on Indicator, Branch on Bit.

Possible saving of transistors 800 - 1300.

14a. Simplify the instruction set by eliminating costly instructions such as Execute.

14b. Eliminate Indirect Execute } 500

14c. Eliminate Store Address 100

14d. Eliminate Rename 200

14e. Eliminate Load Indirect 300

14f. Eliminate Geometric Load 500

Possible saving of transistors 1600.

15. Simplify the instruction set by eliminating, not needed, redundant or little used instruction.

F P Load

a1- Eliminate F. P. multiply
a2- Eliminate F. P. divide
a3- Eliminate F. P. Add Magnitude
a4- Eliminate F. P. Add
b1- Eliminate Load Index Dir. - 100
b2- Eliminate Store Index Dir. - 100
c1- Eliminate Load Refill Dir. - 200

CHECK LIST FOR SIMPLIFYING SIGMA (cont.)

c2- Eliminate Store Refill Dir. 100
d1- Eliminate Compare Count Dir. 100
d2- Eliminate Compare Count Imm. 100
e - Eliminate Compare Value Negative Imm. 50
f1- Eliminate Interchange Augment
f2- Eliminate Interchange Augment to Memory
g - Eliminate Continued Compare

Possible saving of transistors _____.

* Eliminate or Make an Optional Feature.

optional features in Hughes are bad, and 1620 SHARE-type compatibility,